

USER MANUAL

Strata

A terrain painter for HD-2D maps in Unity. Paint levels, ground, walls, edges and stairs in the Scene view, then bake them to a plain prefab.

Version 0.6.5

Unity 6 · Universal Render Pipeline

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1 Quick Dive

The quickest way to start: make a terrain, drag your textures onto the Strata window, and paint. This chapter gets you there in a few minutes. The chapters after it explain every feature in detail.

Requirements

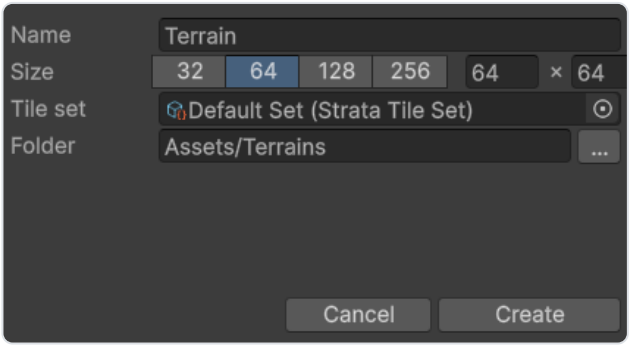
- Unity 6 (6000.0 or newer).
- The Universal Render Pipeline, with a URP asset assigned in **Project Settings** › **Graphics** and **Quality**.

Strata is an editor tool. Everything it needs lives in `Assets/Strata`. The terrains it makes are plain meshes and materials, so your game never depends on Strata's editor code.

Your first terrain

1 Make a terrain

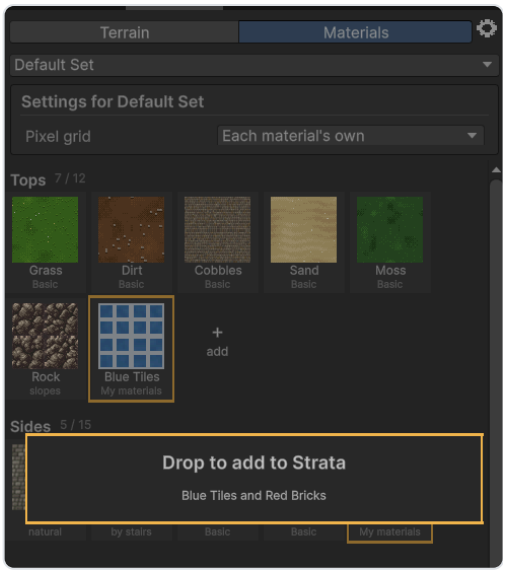
Choose **Tools** › **Strata** › **New Terrain...**. Give it a name, pick a size (64 × 64 metres is a good start) and press **Create**. Strata places the terrain in the open scene and switches the Scene view to the Strata Terrain tool.



The New Terrain dialog. The Default Set holds Strata's ready-made materials.

2 Bring in your textures

Open the Strata window (**Tools** › **Strata** › **Terrain Editor**) and drag textures onto it: one file, many files, a whole folder, or files straight from Explorer or Finder. Strata turns them into materials and matches their normal and roughness maps by name. Skip this step to paint with the materials that come with Strata.



Dropping two textures onto the Strata window.

3 Paint

In the Scene view, the Strata panel shows five brushes. **Level** raises and digs ground, and walls form wherever two heights meet. **Paint** paints the ground. **Stairs** connects two heights, **Shape** makes hills, valleys and ramps, and **Revert** undoes an area back to the original.



Painting ground with the Paint brush. The Strata panel floats in the Scene view while the tool is active.

4 Bake

Press **Bake** in the panel. It also happens by itself when you press Play or save the scene. The terrain is now a regular prefab with meshes and colliders, ready for your game.

> **Tip** Open `Assets/Strata/Examples/Strata Sandbox` to see every brush, material and stair type in one small scene.

2 Core Concepts

A few ideas explain almost everything Strata does.

A grid of one-metre cells

A Strata terrain is a grid of square cells, each one metre wide. Its width runs east to west and its depth runs north to south. Brushes work on whole cells, except the Paint brush, which paints ground smoothly at half-metre resolution.

Natural ground and levels

A new terrain is flat **natural ground**. Natural ground can slope: the Shape brush makes hills, valleys, craters and ramps out of it. The Level brush instead sets cells to a flat **level**, raised above or dug below the ground around them. Levels step by a set amount, one metre by default.

Walls and edges form by themselves

Wherever two heights meet, Strata builds a **wall** between them and puts an **edge** along its top, like a grass overhang or a row of coping stones. You never draw walls: you choose the style the Level brush gives them, and the terrain does the rest. A wall is never left floating, and no floor hangs in the air.

Four kinds of material

Kind	What it covers	Examples in Strata's Basic set
Tops	The ground: every walkable surface.	Grass, Dirt, Cobbles, Sand, Moss, Rock
Sides	The walls between two heights.	Stone, Brick, Earth, Marble
Edges	The trim along the top of a wall.	Turf, Coping, Field Stones
Stair types	How a stair is built.	Stone, Timber, Dirt, Flat Stone, Brick, Marble

A terrain paints from one **tile set**, which lists the tops, sides, edges and stair types it can use. See chapter 7.

Pixel art at sixteen texels a metre

Strata is made for pixel art. Ground and walls are drawn at sixteen texels per metre, so a 256 × 256 texture covers 16 × 16 metres. Borders between ground materials follow each material's pixel grid, so blends stay crisp instead of smearing.

Data, live meshes and baking

Each terrain keeps its own data file (`.strata`). While you paint, Strata updates the meshes right away. **Baking** welds the meshes, refreshes the colliders and saves the data. The result is a prefab made of 64-metre chunks, each with ground, wall and stair meshes. See chapter 8.

3 Creating a Terrain

Open the New Terrain dialog in any of these ways:

- **Tools** › **Strata** › **New Terrain...**
- **GameObject** › **Strata** › **Terrain**, or right-click in the Hierarchy and choose **Strata** › **Terrain**
- The **New terrain...** button in the Strata window, or in the Strata panel when no terrain is open

Field	Meaning
Name	The terrain's name. It also names its folder, data file and prefab.
Size	Cells (metres) east to west and north to south. Use the presets (32, 64, 128, 256) or type any size from 8 to 2048.
Tile set	What the terrain paints with. It defaults to the Default Set (see 7.6). You can change it later in the Strata window.
Folder	Where the terrain's own folder is created. The default is `Assets/Terrains`.

Press **Create**. Strata creates a folder for the terrain, places its prefab in the open scene, opens it for editing and selects the Strata Terrain tool.

What a terrain is made of

File	Purpose
`Name.strata`	The terrain's data: heights, walls, stairs and ground paint. This is the file that matters.
`Name.original.strata`	The terrain as it was first made. The Revert brush paints back to it.
`Name.prefab`	The baked terrain, the object you place in scenes.
`Meshes`, `Materials`, `Textures`	The chunk meshes, the terrain's ground and wall materials, and its paint and wall masks.

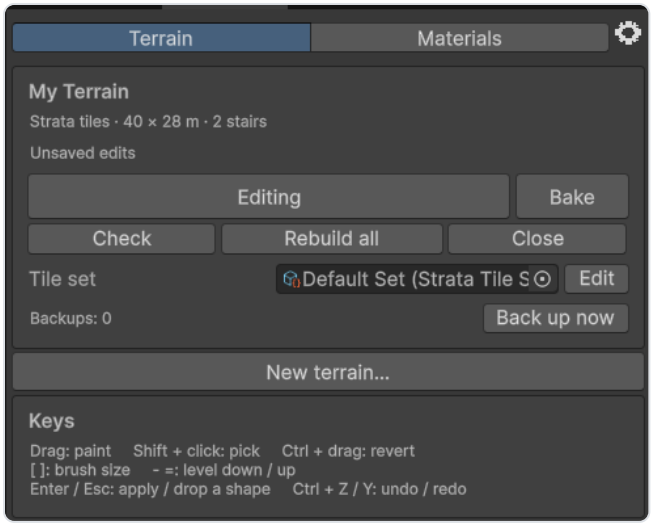
> **Note** When Strata saves, it keeps the previous data file next to the new one as `Name.strata~`, in case a save is interrupted. Unity ignores files ending in a tilde.

Opening a terrain

Strata edits one terrain at a time. Selecting a terrain in the scene and activating the Strata Terrain tool opens it. The Strata window also lists every terrain in the project with an **Open** button. A terrain can only be edited while its prefab is in the open scene. If it isn't, Strata says so: drag the prefab in, or open the scene that holds it.

4 The Strata Window

Open it with **Tools** › **Strata** › **Terrain Editor**. It has two tabs, **Terrain** and **Materials**, a gear button that opens Strata's preferences, and the **On this terrain** panel along its bottom. You can also drop textures and models anywhere on it (see 7.5).



The Terrain tab with a terrain open.

The Terrain tab

With a terrain open, the tab shows its name, format, size and number of stairs, and whether it has unsaved edits.

Edit	Activates the Strata Terrain tool in the Scene view.
Bake	Welds, refreshes colliders and saves (chapter 8).
Check	Compares the meshes, mask and paint with the data and reports any difference.
Rebuild all	Rebuilds every chunk from the data.
Close	Stops editing this terrain.
Tile set	The tile set the terrain paints from. **Edit** opens it in the Materials tab.
Backups	**Back up now** and **Restore...** (chapter 9).

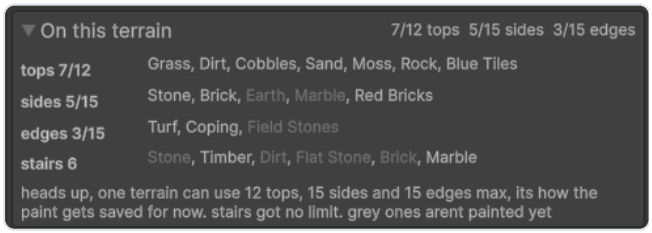
With no terrain open, the tab lists the project's terrains instead, each with an **Open** button. A **Keys** box at the end lists the editing shortcuts.

Changing a terrain's tile set

Pick another tile set in the **Tile set** field. Cells remember positions in the lists, not specific materials. After the switch, a cell painted with the third top shows the new set's third top. This makes it easy to try a whole new look on the same terrain.

On this terrain

The fold-out at the bottom of the window shows how much of the tile set the open terrain uses, as used / maximum for tops, sides and edges. Items the terrain hasn't painted yet are greyed out. The counts turn orange when two or fewer slots are left and red when a kind is full (see 7.1).



The On this terrain panel.

5 Editing in the Scene View

Editing happens with the **Strata Terrain** tool, which appears in the Scene view's tools overlay. Activate it from there, with **Tools** › **Strata** › **Edit Terrain**, or with **Edit** in the Strata window. While it is active, the **Strata panel** floats in the Scene view.

The Strata panel

- **Brush bar:** Level, Paint, Stairs, Shape and Revert. Hover a button for a reminder of what it does.
- **Options:** the settings of the current brush, described in chapter 6.
- **Footer:** the terrain's name and whether it is saved, the **Bake** button, a button for the Materials tab, and a gear for the preferences.

As you move the mouse, Strata previews the brush on the terrain. Level previews the cells and the height they will take, Stairs previews the flight you are dragging, and labels show heights and sizes. You can turn the labels off in the preferences.

Mouse and keys

Input	Action
Left-drag	Apply the current brush.
`Shift` + click	Pick from the terrain. Level picks the cell's height, wall style and edge, plus its ground if Top is on. Paint picks the ground layer under the cursor.
`Ctrl` + drag	Revert with the current brush, back to how the terrain was first made. Use `Cmd` on macOS.
`Ctrl` + click (Stairs)	Remove the flight of stairs under the cursor.
`[` and `]`	Smaller or bigger brush. For the Paint brush this changes the radius.
`-` and `=`	Level down or up by one step. `Page Down`, `Page Up` and the keypad `-` `+` work too.
`Enter` / `Esc`	Apply or drop a Shape selection. `Esc` also cancels a stair you are dragging.
`Ctrl` + `Z` / `Y`	Undo or redo. Each stroke is one step.
`Alt` + drag, right or middle mouse	Navigate the Scene view as usual.

> **Tip** Strata keeps your brush settings between sessions, so the panel opens the way you left it.

6 The Brushes

6.1 Level

The Level brush shapes the terrain into flat levels: terraces, plateaus, sunken courts and pools. Three toggles at the top of its options choose what a stroke does, in any mix. At least one is always on.

Toggle	What a stroke does
Height	Raises or digs the cells to the chosen level. Walls form wherever the new height meets a different one.
Top	Paints the cells' ground with the chosen ground layer, as whole cells with crisp edges.
Sides	Sets the style of the cells' walls (**Sides**) and of the edge along their top (**Top edge**).



Raising a terrace with the Level brush. Walls and edges form on their own.

Useful combinations

- **Height + Sides** (the default): build terraces with the walls and edges you choose.
- **Sides alone**: restyle existing walls, for example turn a stone wall into brick, without touching any height.
- **Top alone**: paint ground in whole cells, for paths and plazas with exact borders.



The Level brush with only Sides on restyles walls and their edge.

Options

Option	Meaning
Level	The target height. Use − and + (or the keys `-` `=`) to move one step, and pick the step from the menu: 0.25, 0.5, 1, 2 or 4 metres. The label shows both the level number and the height in metres.
Sides, Top edge	The wall style and the edge style, from the terrain's tile set.
Ground	With Top on: the ground layer to paint.
Size	The brush size in cells. The slider goes to 64, and the number field allows larger sizes, up to the limit set in the preferences.
Brush	Square or Round.
Snap	How the brush moves: **Cell** (cell by cell), ¼ or ½ (a quarter or half of the brush at a time), **Block** (a whole brush at a time, so squares tile neatly) or **Custom** (a grid of any number of cells).

How walls are styled

A wall takes the style of the cells you painted: raised cells style the walls below them, and dug cells style the walls around them. Where natural ground forms a wall, Strata uses the tile set's first side and first edge. Walls beside stairs use the second side and second edge. The Materials tab labels these items *natural* and *by stairs*.

> **Tip** With **Meet natural ground** on (the default, in the preferences), natural ground right beside a new level comes up or down to meet it when the difference is small. This avoids walls only a few centimetres tall.

6.2 Paint

The Paint brush paints ground layers with a soft, round brush. It works at half-metre resolution, so you can paint organic shapes like patches of moss, worn trails and sand around a pond.

Option	Meaning
Ground	The layers of the tile set's tops, shown as thumbnails. Click one to paint with it.
Radius	The brush radius, from 0.5 to 16 metres. `[` and `]` change it too.
Strength	How much each pass adds, from 0.05 to 1.
Edge	How much of the brush paints at full strength, from 0 (soft all the way) to 1 (hard).

Where two ground layers meet, the border follows the pixel grid and the settings of each top (see 7.3). The top with the higher priority lies over the other. Layers of different families, such as turf against paving, get a lit lip and a small shadow. `Shift` + click picks the layer under the cursor, and `Ctrl` + drag paints back to the original ground.

6.3 Stairs

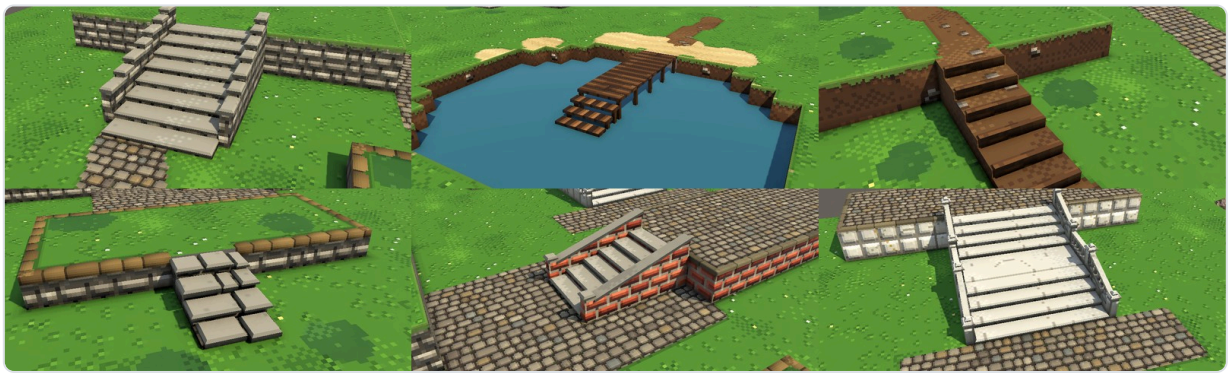
Stairs connect two heights. **Drag from the foot of the stair to its top.** The rectangle you drag is the stair's footprint, and its longer side sets the direction it climbs.



Dragging a marble stair up to a terrace.

Option	Meaning
Type	The stair types of the tile set (see 7.4).
Sides	A toggle for the type's sides, labelled after them: **Cheek walls**, **Balustrade** or **Wing walls**. Types without sides don't show it.
Height	**Auto** climbs from the ground before the foot to the ground beyond the head. **Custom** climbs a fixed height from the foot, in steps of 0.25 m, and can end against a wall.

- Every step rises 0.25 metres, so a two-metre climb has eight steps.
- When both ends are at the same height, the drag makes a **landing**: a flat platform built like the stair. Chain flights and landings to make switchbacks.
- Drawing a stair over part of another one trims or replaces the old one.
- Ctrl + click removes a flight.



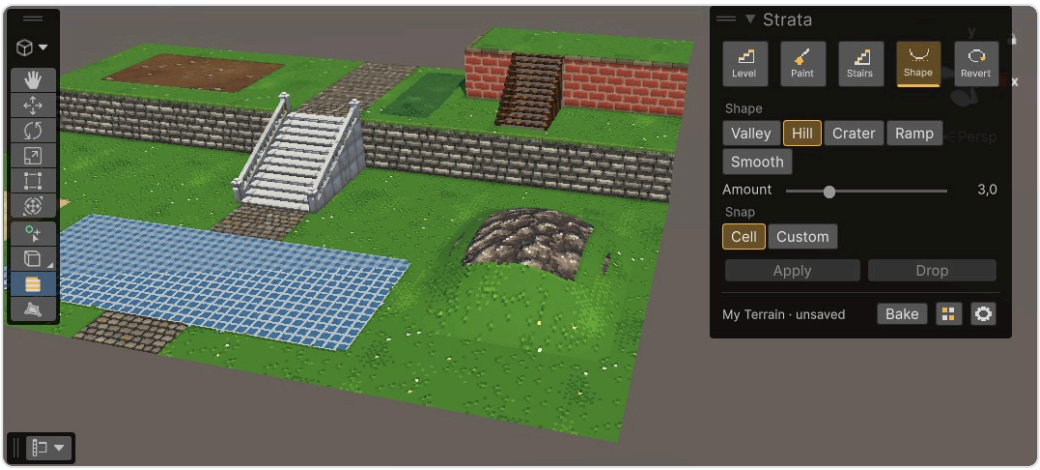
The six stair types that come with Strata. Top row: Stone, Timber, Dirt. Bottom row: Flat Stone, Brick, Marble.

6.4 Shape

The Shape brush sculpts natural ground over a rectangle. Drag to select the area, adjust the options, then press **Apply** (Enter) or **Drop** (Esc).

Shape	Result
Valley	A chasm running along the long side of the selection.
Hill	A rounded hill.
Crater	A bowl with a raised lip.
Ramp	A slope from one end's height to the other's.
Smooth	Smooths what is already there.

Amount is the depth or height in metres (0 to 12). For Smooth, it sets the number of smoothing passes. **Snap** can make the selection follow a grid of a custom number of cells. Natural slopes that are too steep for ground paint show the tile set's slope top, which is Rock in the Basic set.



A hill made with the Shape brush. Its steep sides show the slope top.

6.5 Revert

The Revert brush paints the terrain back to how it was when it was made. Two toggles choose what comes back: **Heights and walls**, and **Ground paint**. Size, shape and snap work as for the Level brush. You can also revert with any brush by holding `Ctrl` while you drag.

7 Materials and Tile Sets

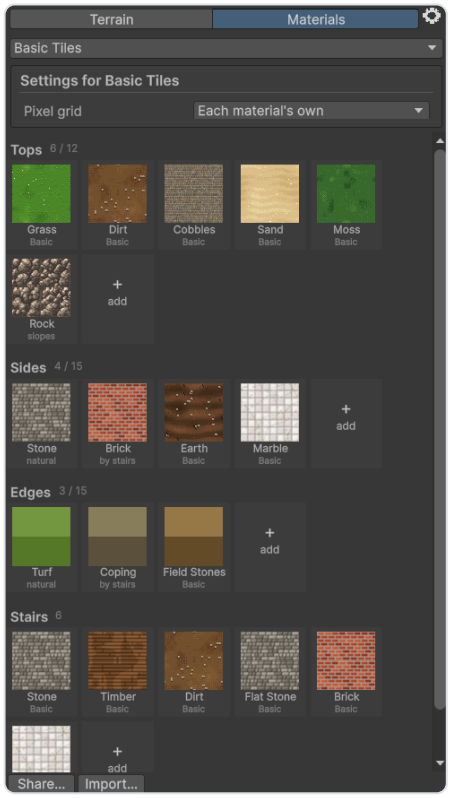
A **Strata material** is an asset holding a material's textures and settings. A **tile set** lists the materials and stair types a terrain can paint with. You manage both in the Materials tab of the Strata window.

7.1 Kinds of material and their limits

Kind	Per tile set	Used for
Tops	up to 12	Ground surfaces
Sides	up to 15	Walls
Edges	up to 15	The trim along the top of walls
Stair types	up to 256	Stairs and landings

The limits come from how a terrain stores its paint. When a kind is full, Strata still creates the new material but doesn't add it to the set, and the On this terrain panel opens to show what the terrain uses. To make room, remove an item the terrain doesn't use, or give that look its own tile set.

7.2 The Materials tab



A tile set in the Materials tab.

The menu at the top chooses what the tab shows: any tile set in the project, **All materials** (every Strata material and stair type in the project), or **New tile set...**, which makes a new set with the same items as the current one.

Below it, **Settings for** shows the selected item, or the tile set itself when nothing is selected. For a tile set, the setting is the **Pixel grid**. By default each material uses its own grid. You can also force one resolution for the whole set, from 512 down to 16 pixels for a 16-metre tile.

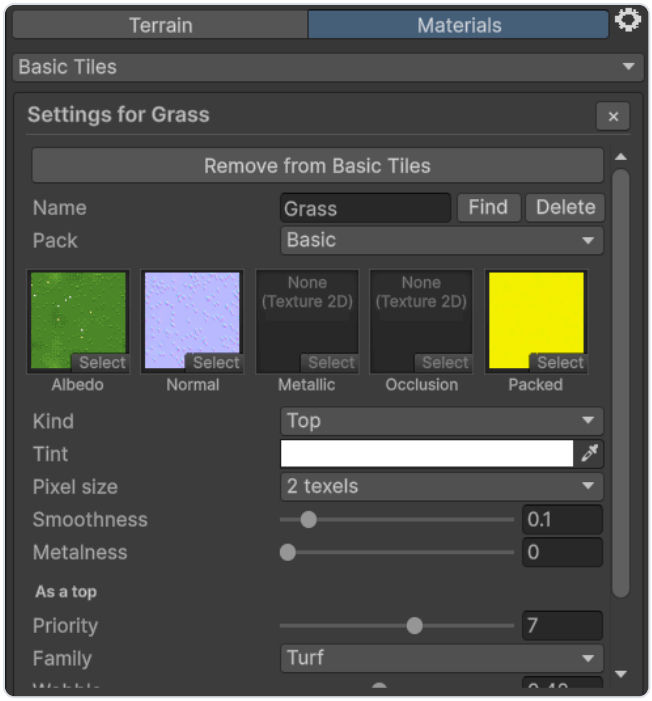
Then come the lists: Tops, Sides, Edges and Stairs, each with its count and limit. Under each thumbnail is the item's pack, or the role it plays: *natural*, *by stairs* or *slopes*. The **add** tile adds a new item or an existing one. New and dropped items blink so you can see where they went.

When an item is selected, its panel offers **Remove from** the set, **Find** (show it in the Project window) and **Delete**.

Share... and **Import...** at the bottom export and import materials as `.unitypackage` files (see 7.7).

Note Removing an item from a set also updates every terrain that uses the set. Cells painted with it show the set's first remaining item of that kind instead (a removed edge simply disappears if no other is left), and everything else keeps its look. Each affected terrain is backed up first, and this can't be undone.

7.3 Material settings



A top selected in the Materials tab.

Maps

Albedo	The colour. A 256 × 256 texture covers 16 × 16 metres.
Normal	A Unity normal map or a plain RGB normal texture.
Metallic	R: metallic, A: smoothness.
Occlusion	Ambient occlusion, greyscale.
Packed	R: occlusion, G: roughness, B: metallic. When set, it replaces the Metallic and Occlusion maps.

General

Kind	Where the material is listed: top, side, edge, or any mix.
Tint	Multiplies the albedo.
Pixel size	How many texels one pixel of the art covers: 1, 2, 4, 8 or 16, or found from the art automatically. Borders and blends follow it.
Smoothness	Used without a metallic map. With one, it scales the map's alpha.
Metalness	Used without a metallic map.

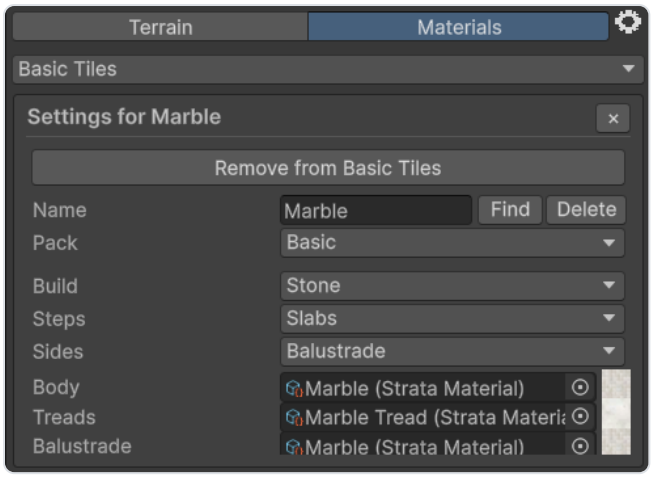
As a top

Priority	Where two tops meet, the higher priority lies on top and wins the border (0 to 12).
Family	Soil, Turf, Paving or Rock. Tops of different families get a lit lip and a shadow where they meet.
Wobble	How much the border wanders (0 to 1).
Ragged	How ragged the border is, texel by texel (0 to 0.3).
On steep slopes	Shows this top wherever natural ground is too steep for paint. One top per set at most.

As an edge

Pattern	**Overhang**: a fringe hanging over the wall, like turf. **Slabs**: cut slabs, like coping. **Stones**: rounded stones.
Use the albedo	Draw the edge with the material's texture. When off, it uses the two colours below.
Dark, Light	For a fringe, the part hanging down the wall and the sunlit lip on the ground. For stones, the joints and darker stones, and the lighter ones.

7.4 Stair types



The Marble stair type.

A stair type decides how a stair is built. Its **Build** is one of three:

- **Stone**: built from Strata materials. Choose the **Steps** and the **Sides**, then the material for each part.
- **Timber**: planks on posts. Each post reaches down to the ground beneath, so timber stairs and landings work as jetties and piers over water.
- **Models**: your own step models (see below).

Steps	**Slabs**: one bevelled slab per step. **Flagstones**: loose stone slabs of different sizes. **Logs**: earth steps edged with timber.
Sides	**Cheek walls**, **Balustrade** (plinth, balusters, rail and newel posts), **Wing walls** (sloped walls with a coping and piers) or **None**.
Parts	Body (the stepped mass), the treads (slabs or timber) and the sides. Timber stairs use Planks and Posts.

The six stair types in Basic Tiles

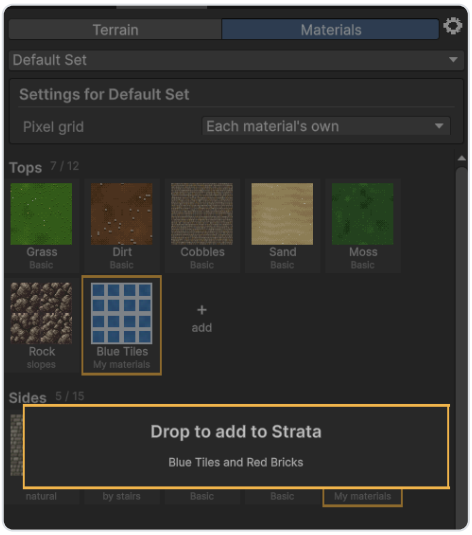
Type	Build
Stone	Stone slabs with cheek walls topped by slabs.
Timber	Planks on posts, for decks, jetties and steps into water.
Dirt	Earth steps, each edged with a squared timber.
Flat Stone	Loose flagstones of varied size, with no sides.
Brick	Slab steps between sloped brick wing walls with piers.
Marble	Marble slabs with a balustrade and newel posts.

Step models

With **Models**, Strata repeats your **Step** model up the flight, fitted to the size of each step. Model it so +Z climbs and +Y is up, or use **Turn** to rotate it in quarter turns. Optional **Bottom step** and **Top step** models replace the first and last step. With **Solid** on, Strata stacks steps under each step down to the foot, so the stair has no hollow underside.

7.5 Drag and drop

Drop textures, folders, Unity materials or models anywhere on the Strata window, from the Project window or straight from Explorer or Finder. Files from outside the project are copied in. Strata groups the files by name, makes a material for each group and adds it to the tile set shown in the Materials tab.



Dropping textures onto the Strata window.

How the name decides the kind

- **Edges:** names with edge, trim, coping, rim, lip, curb, kerb, border, fringe or overhang.
- **Sides:** names with wall, brick, cliff, rockface, facade, masonry, ashlar, plaster, stucco, bank, crag, rampart, palisade, siding or cladding.
- **Stair types:** names with stair, step, tread or riser. A material becomes a stone stair built from it. A model becomes a step model, and names containing first, bottom or lower (or last, top or upper) set the bottom (or top) step.
- **Tops:** everything else. Strata also guesses a top's family and priority from its name: cobbles and tiles become paving, while sand, gravel and dirt become soil, and so on.

To choose the kind yourself, drop onto a section of the list (Tops, Sides, Edges or Stairs).

How maps are matched

Files that differ only by a suffix are treated as one material: albedo, basecolor, diffuse or color for the colour; normal or nrm for normals; metallic, smoothness, roughness, ao or occlusion; orm or arm for packed maps. Normal maps are set up for you, and roughness is converted into a packed map. Small textures keep sharp pixels.

Where new items go

New materials are saved in `Assets/Strata Materials`, and new stair types in its `Stair Types` folder. If no tile set is chosen (All materials), or the chosen set is read-only, they go into the Default Set.

7.6 Basic Tiles and the Default Set

Basic Tiles ships with Strata in `Assets/Strata/Tiles/Basic`. It holds six tops (Grass, Dirt, Cobbles, Sand, Moss, Rock), four sides (Stone, Brick, Earth, Marble), three edges (Turf, Coping, Field Stones) and the six stair types.

The **Default Set** is your own copy of those items. Strata creates it the first time it is needed, in `Assets/Strata Materials/Default Set.asset`. New terrains paint from it, and dropped materials go into it, so you can add, remove and change items freely.

> **Tip** Keep your own materials and tile sets outside `Assets/Strata`. Updating Strata replaces that folder, but never touches `Assets/Strata Materials` or your terrains.

7.7 Sharing materials

Share... exports the selected item, the whole tile set, or a pack as a `.unitypackage`, together with every texture and model it uses, ready for another project. **Import...** imports one. To keep related items together, make a **Strata Pack** (**Create** › **Strata** › **Pack**) and set each item's **Pack** field. Packs label the items in the Materials tab and group them in the add menu.

The Create menu also makes the other Strata assets directly: **Create** › **Strata** › **Tile Set**, **Material** and **Stair Type**.

7.8 Preparing your own art

- Draw at sixteen texels per metre. A 256 × 256 texture covers 16 metres. Sizes that divide 256 (16, 32, 64 and 128) repeat inside that tile, and larger textures are scaled to fit it.
- Make textures seamless and power-of-two sized.
- For chunky pixel art drawn at a lower resolution, set the material's **Pixel size** (2 means each art pixel covers 2 × 2 texels), or let Strata find it from the art.
- Use OpenGL-style normal maps (green pointing up), which is Unity's convention.
- Walls use the same scale as the ground, so a 256-pixel wall texture covers 16 metres along the wall.

8 Baking and the Output

While you edit, Strata rebuilds the parts of the terrain you touch right away. **Baking** finishes the job: it welds the meshes, refreshes the colliders and saves the terrain's data. Press **Bake** in the Strata panel or the Strata window. By default Strata also bakes before entering Play mode and when you save the scene. Both can be turned off in the preferences.

The baked prefab

A terrain's prefab is made of chunks of 64 × 64 metres. Each chunk has three children:

Child	Contents
Ground	The walkable surface with its ground paint.
Walls	Every wall and edge.
Stairs	Every stair and landing, with one material per part.

Each part has a MeshRenderer that casts and receives shadows, and a MeshCollider. The chunks are marked static for batching, occlusion and reflection probes.

At runtime

The baked terrain is ordinary Unity meshes and materials, and your game does not use any of Strata's editor code. Its materials use Strata's shaders (`Strata/TileGround` , `Strata/TileWalls` and `Strata/TileLit`), which Unity includes in builds automatically. Keep `Assets/Strata/Shaders` in any project that uses baked terrains.

Check and Rebuild all

Check compares the terrain's meshes, wall mask and ground paint with its data and reports any difference. **Rebuild all** rebuilds every chunk from the data. Use them if a terrain was edited outside Strata or looks out of date.

9 Undo, Backups and Recovery

Undo

Every stroke, stair and shape is one step in Unity's undo history. Use Ctrl + Z and Ctrl + Y as usual.

Backups

Strata backs terrains up to `StrataBackups/<terrain name>` , next to your project's Assets folder. A bake makes a backup at most once every ten minutes, and Strata keeps the newest thirty for each terrain. Both numbers can be changed in the preferences. In the Strata window, **Back up now** makes one immediately, and **Restore...** lists the

backups. Restoring first backs up the terrain as it is, so it can always be undone. A backup can only be restored onto a terrain of the same size.

Recovering after a crash

Strata keeps unsaved edits in the project's Library folder as you work. If Unity closes before you bake, the next time you open the terrain the Strata window offers **Restore those edits** or **Discard them**.

Safe saving

Strata writes each save to a temporary file and checks it before replacing the terrain's data. The previous version stays next to it as `.strata~`.

10 Preferences

Open them with **Edit** › **Preferences** › **Strata** (**Unity** › **Settings** › **Strata** on macOS), or with the gear button in the Strata window or panel. **Reset to defaults** restores everything.

Setting	Default	Meaning
Baking		
Bake before Play	On	Bakes the open terrain when entering Play mode.
Bake when the scene is saved	On	Bakes the open terrain when its scene is saved.
Backups		
Backups kept per terrain	30	Older backups are deleted (1 to 200).
Minutes between backups	10	A bake backs up at most this often (0 to 120; 0 backs up on every bake).
Brushes		
Largest brush (cells)	256	The largest brush size you can type (8 to 1024).
Meet natural ground	On	Natural ground beside a new level comes level with it when it is within about half a level step (0.45 m at most), so no thin wall is left.
Labels in the Scene view	On	Shows heights and sizes next to the brush.
Experimental		
Objects follow the ground	Off	Props, trees, buildings and characters standing on ground that a stroke raises or lowers move with it. The moves undo with the stroke.
Standing within (m)	0.3	How close an object's base must be to the ground to count as standing on it.
Largest object (m)	24	Wider objects, such as water or whole groups, stay put and their parts are checked one by one. The width comes from colliders where there are any.

11 The Sandbox Example

Open `Assets/Strata/Examples/Strata Sandbox`. Its 64 × 64 metre terrain shows every material and stair type in a place that suits it:



The Sandbox from above.

- **Stone plateau:** two stone flights with cheek walls, and a cobbled path.
- **Palace terrace:** marble flights and a landing with balustrades, on a marble-walled terrace.
- **Town street:** brick steps between wing walls, up to a cobbled quay.
- **Garden:** flagstone steps up a dry-stone terrace edged with field stones.
- **Forest bank:** timber-edged dirt steps on an earth bank, with a trail and moss.
- **Pond:** a timber jetty on posts with steps down into the water.
- **Rock valley:** a valley made with the Shape brush, showing the slope top.

To edit it, open the scene and choose **Tools** › **Strata** › **Edit Terrain**. Strata finds the terrain in the scene and opens it. **Revert** brings any area back to the original flat field.

12 What Strata Adds to Your Project

Location	Contents
`Assets/Strata`	The tool: editor scripts, shaders, Basic Tiles and the Sandbox example. Updating Strata replaces this folder.
`Assets/Strata Materials`	Created when needed. Your materials, the textures you drop in from outside the project, and the Default Set.
Your terrain folders	One folder per terrain, by default under `Assets/Terrains`.
`StrataBackups/`	Terrain backups, next to the Assets folder. They are not part of your build.
`Library/Strata`	Unsaved edits kept for crash recovery.
Scripting define	Strata adds the symbol `STRATA` to the project, so your own code can check whether Strata is installed.
Editor preferences	Brush and window settings, stored per user.

Strata's editor code compiles into its own assembly, `Strata.Editor`, and is never part of a player build.

13 Troubleshooting

Problem	Solution
The terrain is pink or magenta.	The Universal Render Pipeline isn't active. Assign a URP asset in **Project Settings** › **Graphics** and in each Quality level.
The Scene view says *No terrain open*.	Open a terrain from the Strata window, select a terrain in the scene, or make a new one.
This terrain is not in the open scene.	Drag the terrain's prefab into the scene, or open the scene that holds it.
A kind is full.	A tile set holds 12 tops, 15 sides and 15 edges. Remove an item the terrain doesn't use, or use a tile set made for this terrain. The On this terrain panel shows what is in use.
The terrain looks different after a change outside Strata.	Press **Check** to see what differs, then **Rebuild all**.
Painting feels slow.	Turn off **Always Refresh** in the Scene view's effects menu. It redraws the Scene view continuously even when nothing changes.
Something went wrong with a terrain.	Undo, or restore a backup from the Strata window (chapter 9).
Moving a terrain to another project.	Copy the terrain's folder together with its tile set and materials. **Share...** exports a tile set with everything it uses.

14 Credits

The Rock texture in Basic Tiles is made from the Dungeon Crawl Stone Soup tiles, released under CC0 (public domain): `github.com/crawl/tiles`. Thank you to its artists.

All other code, shaders, textures and examples were made for Strata.